

搜索引擎，论文写作

主要单词：

- | | |
|-------------------------------|---------------------------------|
| 1. Boolean algebra | 29. graph theory, |
| 2. search engine | 30. guarantee |
| 3. download | 31. represent' |
| 4. index | 32. hyperlink |
| 5. rank | 33. retrieve |
| 6. foundation | 34. hash table |
| 7. application | 35. theorem |
| 8. binary | 36. odd(奇数) |
| 9. decimal | 37. even(偶数) |
| 10. logical | 38. chapter |
| 11. manipulate | 39. summary |
| 12. illustrate | 40. backbone |
| 13. operand | 41. democratic |
| 14. operator | 42. manuscript(手稿) |
| 15. bachelor | 43. keyword |
| 16. master | 44. abstract |
| 17. thesis/dissertation | 45. acknowledgement(致谢) |
| 18. truth table | 46. reference(参考文献) |
| 19. digital circuit | 47. propose |
| 20. web crawler | 48. algorithm |
| 21. discrete mathematics | 49. prove |
| 22. branch | 50. Proof |
| 23. edge | 51. Natural language processing |
| 24. vertex(复数 vertices) | 52. intuitive |
| 25. depict | 53. mathematics |
| 26. breadth first search(BFS) | 54. mathematician |
| 27. depth first search(DFS) | 55. revise |
| 28. backtrack(回溯) | |

段落节选

1.

First and foremost, this book's intended audience is the general public, not just employees of search companies. To the layperson, understanding the role of mathematics in engineering is more important than explicating every single algorithm hidden behind a product. Furthermore, the bases of technology are twofold: theory and implementation. This book aims to ground the reader in the theories behind innovations, rather than provide their code documentation. A lot of search technology will be out of date in a few years, replaced by newer, better versions, but the theory behind them does not change. Those who continually pursue the latest implementations lead hard lives, because they must renew their knowledge every few years. Only those who understand the nature and essence of search can remain relevant throughout their careers.

2.

There is no simpler number system in the world than binary. Binary has two numbers, 0 and 1. From a purely mathematical standpoint, it makes more sense to count in binary than in decimal. Alas, humans have ten fingers, so evolution and history dictate that we count in base ten, not binary or octal.

3.

Boolean algebra is simple, but it holds great importance in modern mathematics and computer science. Not only did it merge logic and mathematics, but it also provided a new perspective on our world and opened up the digital age. To end this chapter, we borrow some of Newton's words:

Truth is ever to be found in simplicity, and not in the multiplicity and confusion of things.

4.

The origins of graph theory can be traced to Leonhard Euler. In 1736, Euler took a trip to Königsberg, Prussia*, where the locals had a peculiar pastime. Königsberg had seven bridges, as depicted in [Figure 9.1](#). The goal was to walk through each bridge exactly once and return to the original point of departure, but no one had ever succeeded. Euler proved that succeeding was actually *impossible* and published a paper to prove this case. People consider this paper the inception of graph theory, and we will describe Euler's proof in the extended reading.

5.

Now that we understand the basics of graph theory, we can uncover the relationship between graph traversal algorithms and web crawlers. Although the Internet is an enormous, complicated place, it too can be represented as a graph. Websites are vertices, and hyperlinks between them are edges. Most Internet users probably know that behind every clickable URL is a hyperlink target, which will take you to another page on the web. Those links are specified within a website's source code, so if we download a web page, we can use graph search to explore all the outgoing links from that page. Programs that automatically explore the Internet are known as web crawlers, or "bots." The first web crawler was invented by an MIT student, Matthew Gray, in 1993. He named his program the "WWW Wanderer." Modern web crawlers are much more complicated than Gray's contraption, but the underlying ideas have not changed.

6.

Today, most search engines return thousands of results for each query, but which of the thousands would users like to see the most? A search engine's quality is determined by how well it answers this question. Here, we will spend two chapters explicating this challenge. In general, search result rankings depend on two criteria: page quality and content relevance. This chapter introduces methods of ensuring page quality, while the next will focus on relevance to the search keywords.

7. Now , we return to the technical aspect of search engines. The basic recipe is simple: automatically download as many web pages as possible; generate an efficient and useful method of indexing; and sort the web pages in order of relevancy, by some fair and accurate metric.

第一课 cloud computing

主要单词:

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|--------------------------|------------------------------|
| 1. cloud computing, | 27. animation studio, |
| 2. grid computing, | 28. distinguish, |
| 3. concept, | 29. reside, |
| 4. specific technology, | 30. unveil, |
| 5. handheld device, | 31. component, |
| 6. refer to, | 32. architecture, |
| 7. productivity (办公软件), | 33. prototype, |
| 8. document, | 34. leading company, |
| 9. social network site, | 35. laser printer, |
| 10. consumer oriented, | 36. cell phone=mobile phone, |
| 11. high-end, | 37. portable, |
| 12. perform, | 38. home appliance(家用电器), |
| 13. performance, | 39. semiconductor(半导体), |
| 14. outsource (外包) | 40. electronic computer, |
| 15. contract, | 41. touch screen |
| 16. equipment, | 42. desktop computer, |
| 17. staff, | 43. notebook computer, |
| 18. memory(内存), | 44. tablet computer, |
| 19. storage (外存), | 45. ultrabook computer, |
| 20. infrastructure(基础设施) | 46. processor(处理器) |
| 21. cofound, | 47. scalability |
| 22. cofounder, | 48. optimal |
| 23. nominate(提名), | 49. optimization |
| 24. nomination, | 50. minimum |
| 25. patent, | 51. maximum |
| 26. motivation, | |
| 52. customization | |
| 53. manufacture | |
| 54. dictation | |
| 55. battery | |

To this end, the computers would necessarily contain five basic components: a memory, a calculating unit (CPU), and input and output components for human users. The control unit would delve into memory, finding the data, and deal with what it found accordingly. Stored-program architecture was an advancement over and was far more flexible than its predecessor. It has become the prototype of most of its successors, including modern computers which exist to this day.↵

段落节选

It is the idea that consumers use their computer storage, and other computing resources supplied by the cloud, to store and process data on their local devices.

The biggest advantages of cloud computing include that the user has access to an active Internet connection. On the device being used, the data is safe if the device is lost or damaged.

In general, the term cloud computing refers to the use of a “cloud” of servers, typically via the Internet. It took several years to create the supercomputer-level power of modern applications, but it was more typically referred to as mainframe computing. Cloud computing typically refers to accessing a service from a computer, mobile phone, or any other Internet-connected device. One advantage of cloud computing is that apps and data are available anywhere. For example, you use cloud computing capabilities to watch videos, and other media online; use programs and services (e.g., email, etc.); and share ideas, opinions, and content with others.

GORDON MOORE—Intel Cofounder

Moore’s Law is a prediction made in 1965 that the number of transistors and resistors placed on a chip doubling would have a proportional increase in performance. The forecast proved amazingly accurate for 10 years, doubling every two years. A recent breakthrough in chip technology which uses light to imprint circuits on a chip, could double the number of transistors and resistors on a chip every year.

Moore cofounded Intel in 1968 with the cofounder of AMD. His lifelong interest in technology is credited to his father, Binocular Telescope, the world’s most technologically advanced telescope in southeastern Arizona.

第二课

主要单词:

- | | |
|-----------------------|--------------------------------------|
| 1. integrated circuit | 18. giant |
| 2. instrument | 19. premier |
| 3. legacy | 20. embark |
| 4. award | 21. unprecedented |
| 5. come across | 22. chip |
| 6. relevant | 23. conceive |
| 7. document | 24. prototype |
| 8. collaboration | 25. indicator |
| 9. register | 26. reveal |
| 10. forum | 27. augment |
| 11. mentor | 28. pursuit |
| 12. website | 29. artificial intelligence |
| 13. semiconductor | 30. acknowledgement (致谢) |
| 14. principle | 31. interaction |
| 15. transistor | 32. motherboard |
| 16. microprocessor | 33. BIOS (Basic Input/output System) |
| 17. headquarter | |

段落节选

1.

As a pioneer of human-computer interaction, D. Engelbart changed the way computers worked, from specialized machinery that only a trained scientist could use, to a user-friendly tool that almost anyone can use. He invented or contributed to several interactive, user-friendly devices: the computer mouse, windows, computer video conferencing, hypermedia, groupware, E-mail, the Internet and more. In 1964, the first prototype computer mouse was made to use with a graphical user interface, "windows". Douglas received a patent for the wooden shell with two wheels in 1970, describing it in the patent application as an "X-Y position indicator for a display system." "It was nicknamed the mouse because the tail came out in the end," Douglas revealed about his invention.

2. Jack Kilby is credited with being one of the influential people in the world. He was awarded more than 60 patents during his lifetime, but none changed the society as much as the one for his integrated circuit, or microchip, that made the microprocessors possible. His Nobel Prize in physics, awarded in 2000, recognized his part in the invention of the integrated circuit.

3.

Intel

The Intel Corporation is the leading maker of semiconductor chips, and its products are found in a vast majority of personal computers. Headquartered in Santa Clara, Calif., Intel was founded in 1968 as Integrated Electronics Corporation by two giants of the early semiconductor industry, Gordon E. Moore and Robert N. Noyce. Mr. Moore is widely known for the principle called Moore's Law, which asserts that the number of transistors on a chip, and therefore the chip's performance, doubles roughly every two years. Mr. Noyce is a co-inventor of the integrated circuit. Intel's third employee was Andrew S. Grove, who led Intel through its transformation into the premier manufacturer of microprocessors and is now remembered as the company's key business and strategic leader.

While Intel created the first commercial microprocessor chip in 1971, it was not until the success of the personal computer that this became their primary business. Originally known primarily to engineers and technologists, Intel's successful "Intel Inside" advertising campaign of the 1990s made it and its Pentium processor household names. During this period Intel embarked on an unprecedented growth and became the dominant supplier of microprocessors for PCs.

第三课

主要单词:

- | | |
|--------------------------------|------------------------------|
| 1. camera | 21. specialist |
| 2. display | 22. dictate |
| 3. touchpad | 23. headline |
| 4. battery | 24. byline |
| 5. microphone | 25. publish |
| 6. science fiction | 26. beam |
| 7. obstruct | 27. license |
| 8. speech recognition | 28. three dimension printing |
| 9. natural language processing | 29. newscaster |
| 10. native resolution | 30. journalist |
| 11. intuitive | 31. manufacture |
| 12. translation | 32. repairman |
| 13. first-person perspective | 33. distributor |
| 14. snap | 34. horizontal |
| 15. footage | 35. z-axis |
| 16. confirm | 36. customize |
| 17. vibrate | 37. founder |
| 18. feature | 38. foundation(基金会) |
| 19. video conference | 39. album |
| 20. boost | 40. multimedia |

- | | |
|---------------------|------------------|
| 41. manipulate | 46. optimization |
| 42. presentation | 47. dominate |
| 43. embedded system | 48. dominance |
| 44. optimize | 49. proportion |
| 45. optimal | 50. clip out |

段落

1.

Essentially, Google Glass is a camera, display, touchpad, battery and microphone built into spectacle frames so that you can perch a display in your field of vision, film, take pictures, search and translate on the go. ^[1]

2.

Directions become more intuitive (although it sounds like there is no GPS on board so you will have to pair it with your phone), you can view real-time translations or transcriptions of what is being said, and you can scroll through and reply to messages—all on the fly.

3.

ADOBE SYSTEMS—Design Software Leader

Practically all creative professionals involved with art and photography have a copy of Adobe Photoshop on their computer, and the leading computer manufacturers ship their products with a copy of Adobe Reader installed. The worldwide presence of Adobe Systems software attests to the company's success in developing programs that help people communicate effectively.

Charles Geschke and John Warnock founded the company in 1982 and named it after a creek that ran behind Warnock's house in California. Creative Suite CS6 contains the fundamental tools that help photographers, designers, and publishers develop and maintain their documents and Web sites, and it includes Dreamweaver, Flash, Fireworks, Premiere, InDesign, Illustrator, Photoshop, etc.

Recently, Adobe was voted one of the 100 Best Companies to Work For.

4.

3D printers use a variety of very different types of additive manufacturing technologies, but they all share one core thing in common: they create a three dimensional object by building it layer by successive layer, until the entire object is complete. It's much like printing in two dimensions on a sheet of paper, but with an added third dimension: UP, the Z-axis.

5.

BILL GATES—Microsoft Founder

When Bill Gates stepped down from his day-to-day activities at Microsoft in 2008, his action marked the end of an era that shaped the computer world. He remains the company's chairman and advisor, but he now devotes much of his time directing the Bill & Melinda Gates Foundation, a philanthropic organization working to help people worldwide lead healthy, productive lives. His foundation currently is awarding \$3 billion in grants to improve education and graduation rates via technology, with an emphasis on online learning.

6.

The Software Giant—Microsoft

Microsoft, the software giant from Redmond, Wash., was born in 1975 as Micro-Soft, by Bill Gates and Paul Allen, a partnership between two high school friends. When that consumer market exploded with the introduction of the IBM personal computer in 1981, the main beneficiary turned out to be the now mighty Microsoft, which develops, manufactures, licenses and supports a range of software products for computing devices. One of Bill Gates' key visions for the company are "to get a workstation running our software onto every desk and eventually in every home".

7. Embedded system is a specialized computer system that is part of a larger system or machine, designed for a particular kind of application device. An embedded system is some combination of computer hardware and software, either fixed in capability or programmability.